

Math OCR + Embedding Test Pack

A short PDF with mixed math text and rendered equations to test OCR, parsing, chunking, and embedding storage.

What this includes

- Plain-text math expressions and word problems (easy extraction).
- Rendered equation images (to test math OCR).
- A small numeric table (to test table extraction).
- A simple function graph (to test figure handling).

Tip: Run your pipeline end-to-end on this PDF. Verify you get non-empty page text, chunks, embeddings, and successful persistence.

Section 1: Plain-Text Math (Extraction Sanity Check)

1) Integers: Evaluate $18 - 27 + 5$. 2) Fractions: Simplify $(12/30)$ and then compute $(3/5) + (7/10)$. 3) Decimals: Convert 0.375 to a fraction in simplest form. 4) Algebra: Solve for x : $5x - 7 = 18$. 5) Ratios: A recipe uses flour:sugar = 3:2. If flour is 450 g, how much sugar is needed? 6) Percent: A jacket costs 2,000. It is discounted by 15%. What is the sale price? 7) Geometry: A rectangle has length 9 cm and width 4 cm. Find area and perimeter. 8) Word problem: A taxi charges a base fee of 120 plus 35 per kilometer. Write the expression for cost after k kilometers and compute the cost for $k=8$.

Small Table (for extraction)

Student	Quiz 1	Quiz 2	Quiz 3	Total
Amina	8	7	9	24
Bilal	6	9	8	23
Hassan	10	8	7	25
Noor	7	6	10	23

Section 2: Rendered Equations (Math OCR Test)

The following lines are images (not selectable text). Use them to confirm your OCR provider extracts equations reliably.

$$\frac{3}{4} + \frac{5}{8} = \frac{11}{8} = 1\frac{3}{8}$$

$$2x + 3 = 11 \Rightarrow 2x = 8 \Rightarrow x = 4$$

$$\sqrt{50} = \sqrt{25 \cdot 2} = 5\sqrt{2}$$

$$A = \pi r^2$$

$$a^2 + b^2 = c^2$$

More equations:

$$\sum_{k=1}^n k = \frac{n(n+1)}{2}$$

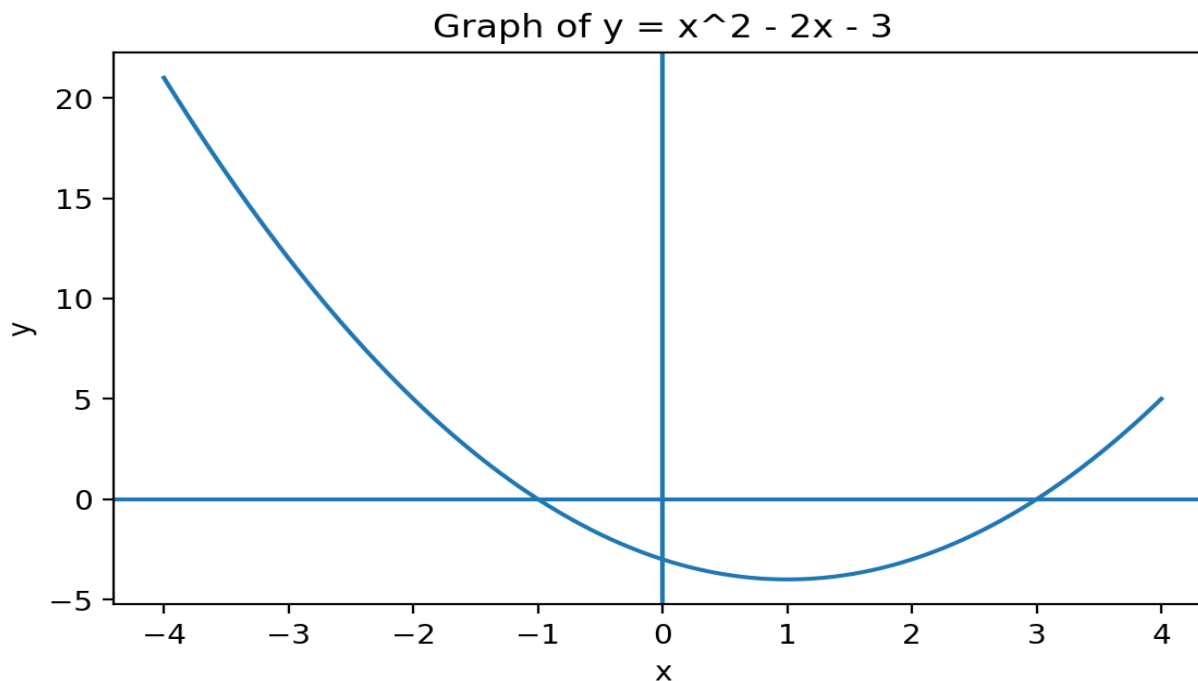
$$\left(x - \frac{1}{2}\right)^2 = x^2 - x + \frac{1}{4}$$

$$\log_{10}(1000) = 3$$

$$\frac{d}{dx}(x^3) = 3x^2$$

Section 3: Figure Handling (Graph)

Below is a simple graph image. Your ingestion should not crash even if you ignore figures.



Section 4: Mixed Symbols (Text)

Units and symbols: 5 cm, 3 m², 9.8 m/s², 60°, ±, !=, <=, >=. Expression: (a + b)² = a² + 2ab + b² Set notation: A = {2, 4, 6, 8}, B = {1, 2, 3, 4}